

Terrain-Conditioned Deep Neural Networks for Next-Day High-Flow Prediction in Sri Lanka

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Abstract—This study investigates MMF-Net, a terrain-conditioned temporal transformer for next-day high-flow prediction at 51 locations in Sri Lanka. The model combines 14-day sequences of 33 hydrometeorological variables with nine static catchment attributes using periodic numerical embeddings, temporal and cross-feature attention, and feature-wise linear modulation. The prediction target is exceedance of a location-specific discharge threshold derived from GloFAS reanalysis, representing a high-flow proxy rather than observed inundation. On the temporal test split, periodic embeddings increase average precision from 0.6083 to 0.7605. The terrain-conditioned five-seed ensemble with binary cross-entropy training and temperature scaling achieves average precision of 0.8355, a Brier score of 0.0080, and expected calibration error of 0.0016. These findings support numerical embeddings and static conditioning for this prediction task. Single-seed component ablations and incomplete radar controls limit broader conclusions; validation against observed floods remains necessary for warning applications.

Index Terms—deep neural networks, temporal transformer, numerical embeddings, terrain conditioning, flood prediction, Sri Lanka

I. INTRODUCTION

Next-day high-flow prediction depends on recent rainfall, antecedent wetness, river discharge, and catchment characteristics. A neural model must capture how these variables interact over time and how local conditions influence the response. Sri Lanka’s contrasting climatic zones provide a setting for studying these relationships across multiple river-basin locations.

Neural networks have shown promise in rainfall–runoff modelling and large-scale flood forecasting [1], [2]. This study examines how numerical representations and static catchment information affect a regional deep neural network. We investigate MMF-Net, which combines periodic numerical embeddings, temporal and cross-feature attention, and feature-wise linear modulation of static attributes. An optional Sentinel-1 branch explores radar fusion.

Using 14-day input histories from 51 locations, we evaluate next-day exceedance of a training-period discharge threshold. Component ablations examine embeddings, attention, and static conditioning; further experiments assess training objectives, ensembling, and probability calibration.

The contribution is an empirical assessment of these established neural components for Sri Lankan high-flow prediction. Because the labels derive from discharge reanalysis, the results

measure prediction of a hydrological proxy. Agreement with observed inundation and suitability for operational warnings require separate validation.

II. RELATED WORK

A. Neural hydrological forecasting

Neural rainfall-runoff models have demonstrated the value of learning hydrological relationships from temporal sequences [1], while recent large-scale work has extended neural prediction to extreme flows in ungauged watersheds [2]. Local work has also applied bidirectional LSTMs to water-level forecasting in Sri Lanka [3]. The present study differs in using daily high-flow classification across multiple Sri Lankan locations with GloFAS-derived reanalysis supervision.

B. Representation and conditioning

Transformers use self-attention to model temporal relationships [4]. Gorishniy *et al.* showed that numerical feature embeddings, including periodic embeddings, can improve deep tabular models [5]. MMF-Net applies periodic embeddings before temporal encoding and uses FiLM conditioning [6] to incorporate static terrain, climatic and network attributes.

River topology can also be incorporated through graph models, but its benefit is not guaranteed. Kirschstein and Sun found no improvement from supplied river topology in their neural forecasting experiments [7]. MMF-Net uses shared weights across locations without inter-location message passing.

C. Evaluation and probability quality

Focal loss addresses class imbalance by reducing the influence of easy examples [8], while deep ensembles and temperature scaling provide approaches to uncertainty estimation and calibration [9], [10]. Precision-recall analysis is particularly useful for imbalanced event prediction [11], while temporal dependence requires evaluation procedures that respect the data structure [12]. These considerations guide the evaluation and interpretation of MMF-Net.

III. STUDY DATA AND PREDICTION TASK

A. Spatial Coverage and Data Provenance

The study covers 51 node locations grouped into 16 hydrological basins/areas in Sri Lanka (including the Colombo

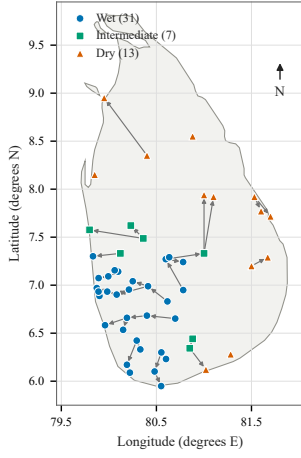


Fig. 1. Study locations and curated links, coloured by climatic zone. Country outline: Natural Earth.

metropolitan area), distributed across the wet (31 nodes), dry (13 nodes), and intermediate (7 nodes) climatic zones. Spatial graph representations comprise 35 curated upstream-to-downstream flow links and 204 spatial links for graph-based architectures (Fig. 1). Gauge-measured discharge is not used directly.

Daily meteorological and soil-wetness attributes are sourced from NASA POWER [13], supplemented by cached Open-Meteo Archive precipitation and GloFAS reanalysis discharge via the Open-Meteo Flood API [14], [15].

The feature space comprises:

- **33 Dynamic Inputs:** Rainfall and antecedent precipitation (11), temperature, humidity, wind, and radiation (7), soil wetness (6), and discharge metrics including rolling means, recent changes, seasonal anomalies, and empirical percentiles (9).
- **9 Static Inputs:** Elevation, climatic-zone indicators (3), network-position indicators (4), and a drainage proxy (log mean-training-discharge).

Heavy-tailed dynamic features are transformed using signed $\log(1 + |x|)$ followed by z-score standardization. Missing normalized values are zero-imputed (representing the training mean).

B. Targets and Episode Definition

Let $Q_{n,t}$ denote reanalysis discharge at location n on day t . The location-specific high-flow threshold τ_n is defined as the 98th empirical percentile of historical daily discharge through 2017:

$$\tau_n = \text{quantile}_{0.98}\{Q_{n,t} : t \leq 2017-12-31\}, \quad (1)$$

$$s_{n,t} = \mathbf{1}[Q_{n,t} \geq \tau_n], \quad y_{n,t}^{(1)} = s_{n,t+1} \quad (2)$$

where $y_{n,t}^{(1)}$ serves as the primary binary target for next-day high-flow prediction.

Auxiliary targets include multi-day exceedance $y_{n,t}^{(h)} = \max_{k=1,\dots,h} s_{n,t+k}$ for $h \in \{2, 3\}$, event onset $y_{n,t}^{(o)} =$

$(1 - s_{n,t})s_{n,t+1}$, and regression heads predicting the next-day discharge percentile and 3-day maximum discharge z-score.

Proxy episodes are defined per node by grouping consecutive positive state days ($s_{n,t} = 1$) separated by at most a one-day gap.

C. Dataset Scope

The primary DNN input sequence is truncated at 31 December 2024. The full catalog contains 1,469 proxy episodes (1,464 starting in 2003–2024) split across training, validation, and testing periods (Fig. 2).

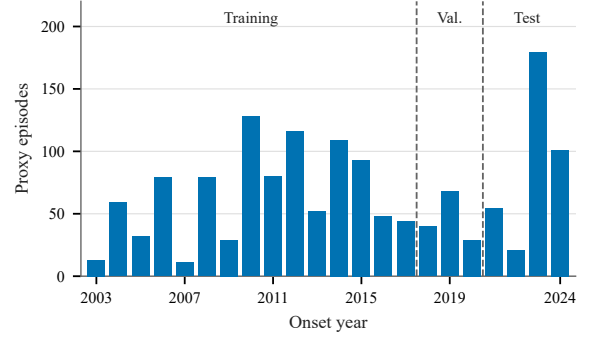


Fig. 2. Annual distribution of node-level proxy episode onsets (2003–2024) across experimental temporal splits.

IV. DEEP NEURAL NETWORK METHOD

A. Numerical and temporal representation

Each prediction uses a 14-day, 33-variable history per node, $\mathbf{X}_{n,t} \in \mathbb{R}^{14 \times 33}$, with nodes processed independently via the batch dimension. Each scalar feature j is embedded using a periodic-linear-ReLU (PLR) representation with trainable sine/cosine frequencies $\mathbf{c}_j \in \mathbb{R}^8$:

$$\mathbf{e}_j(x) = \text{ReLU}(\mathbf{W}_j[\sin(2\pi\mathbf{c}_j x) \parallel \cos(2\pi\mathbf{c}_j x)] + \mathbf{b}_j). \quad (3)$$

Daily embeddings are concatenated, projected to 128 dimensions, and passed (with position and classification tokens) through three pre-LN transformer blocks (4 heads, feed-forward width 256, dropout 0.2), yielding the temporal representation $\mathbf{h}^T$.

B. Cross-feature attention and static conditioning

A parallel branch averages embeddings over time, projects each feature to a 128-dimensional token, and applies a one-layer transformer to obtain $\mathbf{h}^F$. The branches merge as

$$\mathbf{h} = \text{LN}(\mathbf{h}^T + \mathbf{h}^F). \quad (4)$$

A 9-dimensional static vector $\mathbf{s}_n$ supplies the scale and shift vectors γ_n and β_n through a zero-initialised FiLM layer:

$$\tilde{\mathbf{h}} = \text{LN}[\mathbf{h} \odot (1 + \gamma_n) + \beta_n]. \quad (5)$$

A fusion MLP (widths 192, 128) and shared head (widths 256, 128) produce the classification and regression outputs (Fig. 3).

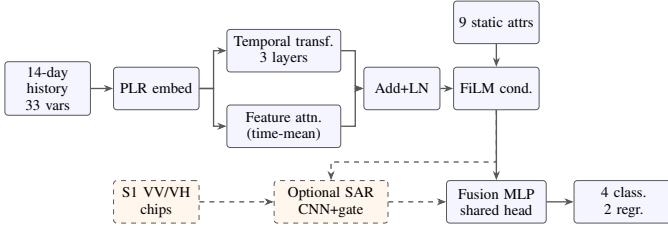


Fig. 3. Model data flow: parallel temporal/feature branches merge and are conditioned on static attributes; optional radar fusion (dashed) feeds the shared head. No graph message passing is used.

C. Optional radar branch

An optional CNN maps Sentinel-1 VV/VH chips to an embedding $\mathbf{v}$. A learned gate (bias initialised to -3) combines it with the tabular state, using a binary presence indicator m and image age a in days:

$$\mathbf{g} = \sigma\{\text{MLP}[\tilde{\mathbf{h}}\|\mathbf{v}\|m\|(a/30)]\}, \quad (6)$$

$$\mathbf{h}^S = \text{LN}(\tilde{\mathbf{h}} + m\mathbf{g} \odot \mathbf{W}_v\mathbf{v}). \quad (7)$$

Here, σ is the sigmoid function and $\mathbf{W}_v$ projects the image embedding; absent frames contribute zero because $m = 0$. A simpler scalar variant appends four image-derived summaries instead. Imagery is available for only 9 of 51 nodes, and no verified pretrained checkpoint exists, so this branch is not a validated pretrained-radar result.

D. Multi-task optimisation and calibrated inference

Training minimises masked BCE across four classification heads (weights 1.0, 0.3, 0.3, 0.5) plus weighted Huber loss over two regressions (0.2); an alternative `focal_conf` loss instead jointly adjusts class weighting, focal focusing ($\alpha = 0.75$, $\gamma = 2$), and label confidence. Optimisation uses AdamW (10^{-3} lr, 10^{-4} decay, cosine schedule with 5 warmup epochs, ≤ 60 epochs, gradient clipping at 1.0, patience-10 early stopping). At inference, five-seed probability averages are temperature calibrated,

$$p_{\text{cal}} = \sigma[\text{logit}(\bar{p})/T], \quad (8)$$

where $\bar{p}$ is the mean predicted probability and $T > 0$ is the temperature. Validation-based selection is described below.

V. EXPERIMENTAL DESIGN

A. Configuration and Ablations

Table I summarises the experimental settings. N0–N3 progressively introduce periodic numerical embeddings, cross-feature attention and FiLM conditioning. N4 evaluates combined focal and label-confidence weighting. N5 and N5_bce use five-seed ensembles with temperature scaling under the weighted objective and binary cross-entropy, respectively. N6 variants explore SAR fusion. Single-seed ablations are interpreted descriptively. Earlier graph-network and conventional baseline results use a different data extent and are reported separately.

TABLE I
EXPERIMENTAL CONFIGURATION.

Setting	Value
Training period	2003–2017
Validation period	2018–2020
Test period	2021–2024
Input window	14 days; 33 dynamic and 9 static features
Transformer layers	3 temporal; 1 cross-feature
Hidden dimension / heads	128 / 4
Batch size	32; SAR CNN: 8
Optimiser / epochs	AdamW; maximum 60
Seeds	N0–N4: 1; ensemble variants: 5

B. Evaluation

Checkpoints are selected using validation episode average precision (AP). Calibration and F1-based threshold selection use validation data only and are frozen for testing. We report node-day AP, Brier score, expected calibration error (ECE) using 15 equal-width bins, probability of detection (POD), false-alarm ratio (FAR), and critical success index (CSI):

$$\begin{aligned} \text{POD} &= \frac{TP}{TP + FN}, \\ \text{FAR} &= \frac{FP}{TP + FP}, \\ \text{CSI} &= \frac{TP}{TP + FP + FN}. \end{aligned} \quad (9)$$

Here, TP , FP and FN denote true positives, false positives and false negatives, respectively.

Episode detection measures the fraction of episodes with an alarm during the preceding seven days. Episode AP uses the maximum pre-onset probability, with flood-free node-weeks as negatives. This retrospective measure does not establish seven-day forecasting skill. Detection rates are interpreted alongside FAR because models use different validation-selected thresholds.

VI. RESULTS

A. Neural representation and static conditioning

Table II reports the recorded MMF-Net results. PLR embeddings raise AP from 0.6083 to 0.7605; feature attention raises it further to 0.7738; static FiLM conditioning raises it to 0.8269. This supports flexible scalar representations and location-dependent conditioning, but the FiLM contrast does not isolate elevation from the other static inputs (drainage proxy, zone, position) fed to the same module. N0–N4 use one seed each, so their differences describe these specific runs, not statistically established component effects.

B. Objective, ensemble and calibration

Replacing BCE with the combined focal/confidence objective moves N3’s AP from 0.8269 to 0.7592 and ECE from 0.0032 to 0.0525. In the ensemble comparison, N5_bce (AP 0.8355, ECE 0.0016) outperforms N5 (0.7846, 0.0043), favouring BCE here without establishing that focal loss is generally unsuitable, or which term in the combined weighting

TABLE II
MAIN DNN RESULTS ON THE TEMPORAL TEST SPLIT.

ID	Configuration	AP	ECE	FAR	Params
N0	Linear embeddings	.6083	.0024	.412	.551
N1	+ PLR	.7605	.0053	.325	.555
N2	+ feature attention	.7738	.0056	.313	.693
N3	+ static FiLM	.8269	.0032	.110	.711
N4	+ focal/confidence	.7592	.0525	.138	.711
N5	N4, ensemble + TS	.7846	.0043	.145	.711
N5_bce	N3, ensemble + TS	.8355	.0016	.120	.711
N6-sc	SAR scalars, focal/conf.	.7284	.0055	.344	.719
N6-gat	SAR CNN, focal/conf.	.8310	.0066	.107	11.967

TS: temperature scaling; focal/conf.: focal/confidence objective. N0–N4: one seed; others: five. Params: trainable millions/member. Bold = largest recorded AP, not significance. Source: model-2 snapshot, 10–11 Aug 2026.

drives the change. The N5_bce ensemble records Brier 0.0080 and CSI 0.561, exceeding N3’s AP by 0.0086. Temperature scaling preserves ranking, so it does not explain the AP gain; separate effects on probability quality are not established. Reliability curves and event-stratified calibration would be more informative than ECE alone but are not recoverable from the available results.

C. Radar fusion and model size

Scalar radar: AP 0.7284. Gated CNN: AP 0.8310, versus 0.7846 (image-free focal/confidence ensemble) and 0.8355 (image-free BCE ensemble) - mixed evidence, since the CNN beats the focal/confidence family but not the strongest image-free result; differing objectives, batch sizes, uncertain pre-training, and limited node coverage prevent attributing the gap to radar alone. N5_bce uses ~ 0.711 M parameters/member vs. 11.967M for N6-gat; a five-member image-free ensemble totals ~ 3.56 M parameters and needs five forward passes. Parameter counts include gradient-enabled weights, so checkpoint freeze status must accompany any size comparison; latency and energy were not measured.

D. Detection at recorded operating points

Under the seven-day rule: N2 detects 0.423 of episodes at FAR 0.313; N3 detects 0.197 at FAR 0.110; N5_bce detects 0.220 at FAR 0.120. Each point reflects both model and validation-selected threshold; these are not points on a shared operating curve and do not establish preference at a fixed false-alarm budget.

E. Historical references and split diagnostics

Discharge-percentile AP (≈ 0.816) indicates substantial persistence in the next-day target; LightGBM reaches 0.8496 on its own historical panel (Table III). These motivate strong references for a matched rerun but do not establish a DNN-vs-tree ordering on an identical test set; comparing graph-based M3 to MMF-Net would also change encoder and panel, not just graph structure. M3 diagnostics show AP 0.704 (temporal split) vs. 0.905 (random split), with seven-day detection dropping from 0.380 to 0.089 - the random protocol interleaves node-days and can remove pre-onset alarm

TABLE III
HISTORICAL REFERENCES (2 AUG 2026); PANEL EXTENT DIFFERS FROM TABLE II. DASHES: UNAVAILABLE.

Reference	AP	Brier	FAR	Detection
Persistence	.590	-	.231	.223
Climatology	.108	-	.882	.515
Discharge percentile	.816	-	.231	.223
LightGBM	.8496	.0076	.080	.161
GRU only (M0)	.598	.0137	.441	.262
TF-STGNN (M3)	.704	.0135	.342	.380

opportunities, so this shows split sensitivity without isolating a causal loss of warning skill or proving memorisation. The Gin holdout (AP 0.840) tests spatial transfer on four nodes (training elsewhere spans later dates) and is not a future-time test; strict spatial evaluation would also require revisiting held-out-node preprocessing statistics and graph access.

VII. PRACTICAL INTERPRETATION AND LIMITATIONS

A. Operational Relevance

MMF-Net could support river monitoring by presenting high-flow probabilities alongside observed water levels and rainfall. However, reanalysis-derived targets represent a hydrological proxy, and operational warning capability has not been validated. Effective lead time depends on actual input availability, including publication delays and radar acquisition times.

B. Validation and Reproducibility

The high-flow proxy activates in seven of ten documented flood windows during 2003–2024, missing the May 2003, June 2014 and May 2017 events. This limited label-agreement check does not measure DNN warning accuracy. Independent gauge and inundation observations are needed to assess missed events and false alarms.

Future targets can cross temporal split boundaries and must be excluded in a controlled rerun. Exact sample counts, run configurations and prediction archives are also required for reproducibility. Since the test set informed model development, confirmation requires an independent evaluation period.

C. Further Evaluation

Repeated DNN ablations should hold data, seeds and training settings fixed to isolate the effects of embeddings, FiLM, graph structure and radar fusion. Operational evaluation should use validation-selected thresholds under a stated false-alarm constraint, report realised test FAR, and assess horizon-aligned event detection. Calibration and uncertainty estimates should account for temporal and cross-node dependence.

VIII. CONCLUSION

This study examines a terrain-conditioned deep neural network for next-day high-flow prediction at Sri Lankan river-basin locations. Recorded ablations show improvements after periodic numerical embeddings and static FiLM conditioning are introduced, while the BCE ensemble attains the highest

recorded MMF-Net AP of 0.8355. Radar results remain exploratory and the available experiments do not isolate the value of graph structure. The findings support further controlled study of neural representation and fusion. Establishing a practical warning benefit requires independent flood observations, issue-time input replay and matched evaluation of onset detection, false alarms and probability calibration.

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